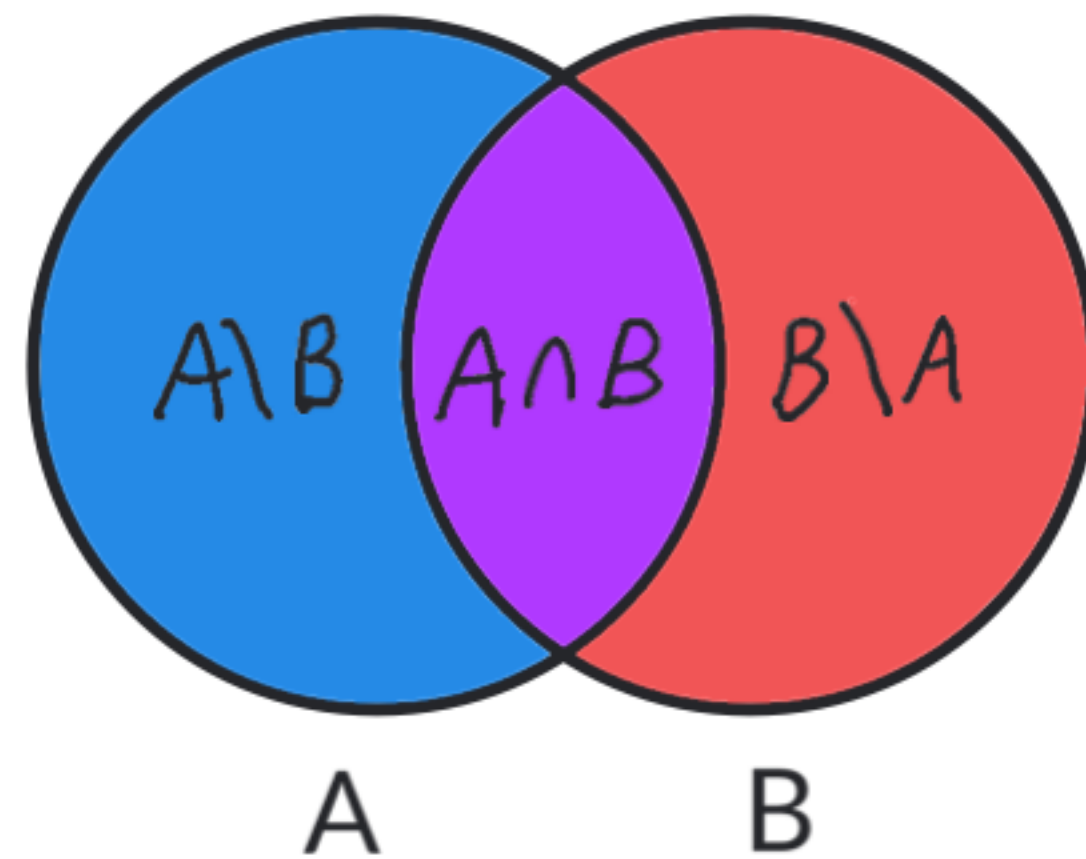


Inclusion-Exclusion Principle



Inclusion-Exclusion Principle:

Combinatorics: $|A \cup B| = |A| + |B| - |A \cap B|$

Probability Theory: $\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$

Example: If we roll 2 fair 6-sided dice, what is the probability that at least one of them is a 6?

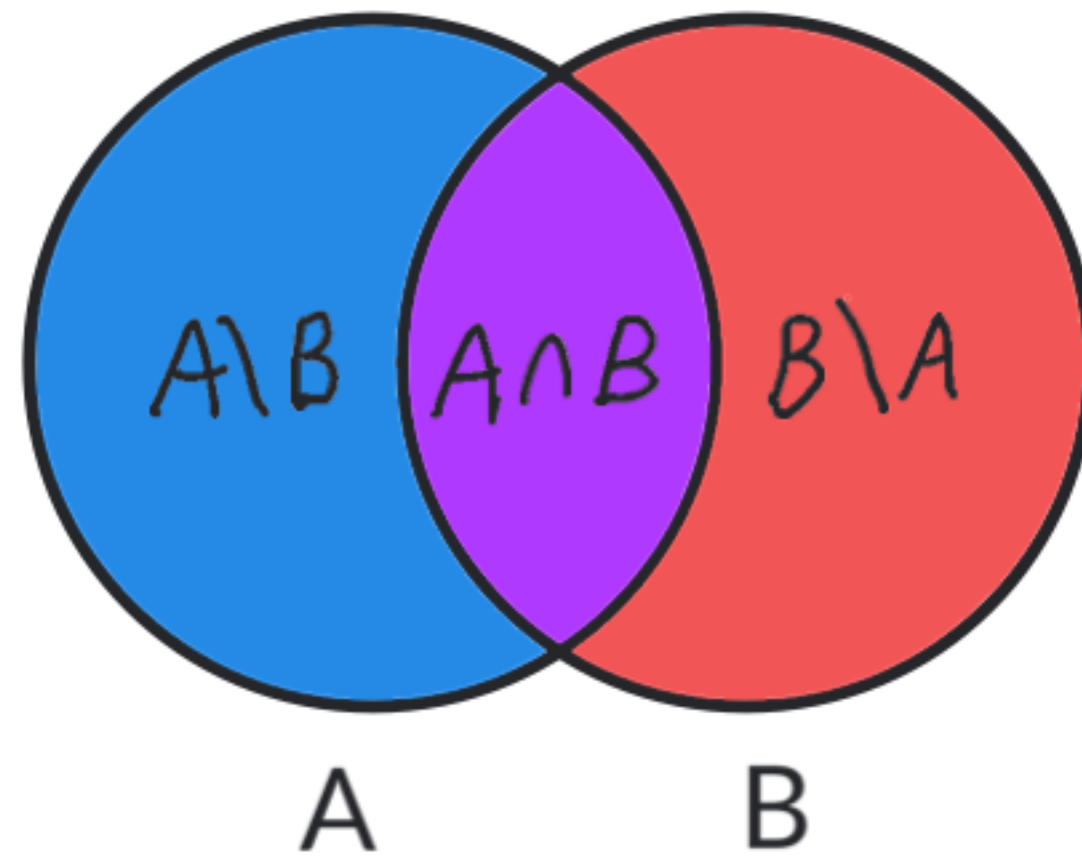
A: The first die is a 6

B: The second die is a 6

$$\Pr(A) = \frac{1}{6}, \quad \Pr(B) = \frac{1}{6}, \quad \text{and} \quad \Pr(A \cap B) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

$$\Pr(A \cup B) = \frac{1}{6} + \frac{1}{6} - \frac{1}{36} = \frac{2}{6} - \frac{1}{36} = \frac{12}{36} - \frac{1}{36} = \frac{11}{36}$$

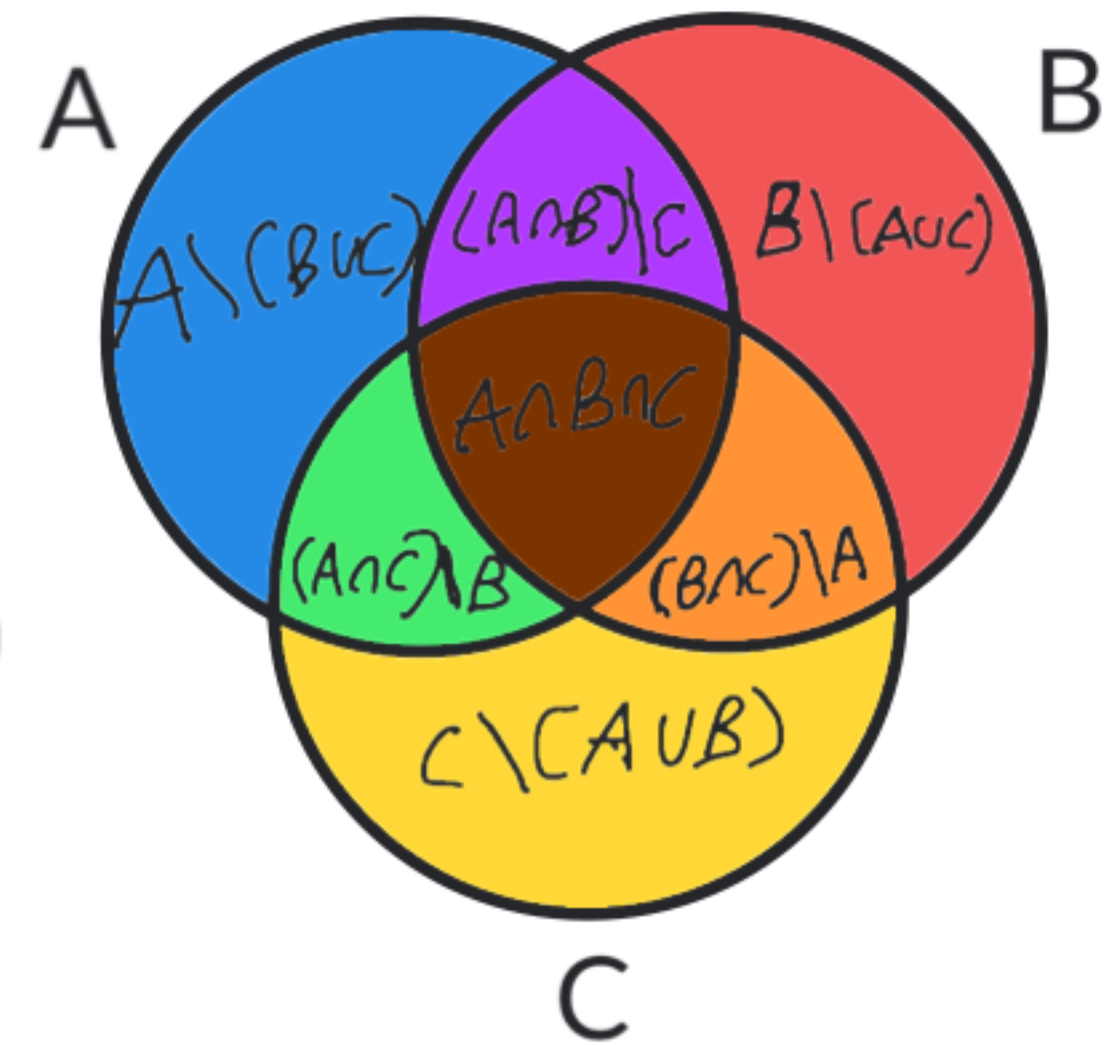
Inclusion-Exclusion Principle Proof



Inclusion-Exclusion Principle for 3 Events

Inclusion-Exclusion Principle
for 3 events:

$$\begin{aligned}Pr(A \cup B \cup C) &= Pr(A) + Pr(B) + Pr(C) \\ &\quad - Pr(A \cap B) - Pr(A \cap C) - Pr(B \cap C) \\ &\quad + Pr(A \cap B \cap C)\end{aligned}$$



$$Pr((A \cup B) \cup C) = Pr(A \cup B) + Pr(C) - Pr((A \cup B) \cap C)$$

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

$$\text{Observe that } (A \cup B) \cap C = (A \cap C) \cup (B \cap C)$$

$$\begin{aligned}-Pr((A \cup B) \cap C) &= -Pr((A \cap C) \cup (B \cap C)) = -Pr(A \cap C) - Pr(B \cap C) \\ &\quad + Pr(A \cap B \cap C)\end{aligned}$$

$$Pr(A \cup B \cup C) = Pr(A) + Pr(B) + Pr(C) - Pr(A \cap B) - Pr(A \cap C) - Pr(B \cap C) + Pr(A \cap B \cap C)$$

Challenge: What happens for k events? Can you prove it?